

Grammar Addendum – slash form + namespace prefix + arrow form (operator mandate, – – ; arrow form – –)

EXTENDS the §11.4.140 action-prefix design with additional equivalences. Every registered action (e.g. BACKGROUND, REMINDER) MUST be invocable in ALL of these EQUIVALENT forms — same action, same expansion, same execution:

#	Form	Example	Notes
1	<code>ACTION_NAME ::</code> <code><rest></code>	<code>BACKGROUND :: Do X</code>	bare <code>::</code> form (no namespace)
2	<code>PRE-</code> <code>FIX::AC-</code> <code>TION_NAME ::</code> <code><rest></code>	<code>DEFAULT::BACKGROUND ::</code> <code>Do X</code>	namespaced <code>::</code> form
3	<code>/ACTION_NAME</code> <code><rest></code>	<code>/BACKGROUND Do X</code>	bare slash form — ONLY if <code>/ACTION_NAME</code> does NOT collide with an existing/built-in slash command of the host agent
4	<code>/PRE-</code> <code>FIX::ACTION_NAME</code> <code><rest></code>	<code>/DEFAULT::BACKGROUND Do</code> <code>X</code>	namespaced slash form — disambiguates from any existing <code>/AC-</code> <code>TION_NAME</code> ; ALWAYS safe
5	<code>ACTION_NAME ----></code> <code><rest></code> (and <code>PRE-</code> <code>FIX::ACTION_NAME</code> <code>----> <rest></code>)	<code>BACKGROUND ----> Do X /</code> <code>DEFAULT::BACKGROUND ---</code> <code>> Do X</code>	arrow form — the <code>----></code> body separator (one ASCII space on each side); a third equi- valent delimiter along- side <code>::</code> and <code>/</code>

Namespace prefix

- `PREFIX` is an action NAMESPACE. The reserved default namespace is `DEFAULT` . Custom namespaces are permitted (e.g. a project/plugin namespace) so the same action can run under the default OR a custom prefix.
- An action is executable WITH or WITHOUT the prefix: `BACKGROUND ::` $\equiv$ `DEFAULT::BACKGROUND ::` $\equiv$ `/BACKGROUND` $\equiv$ `/DEFAULT::BACKGROUND` $\equiv$ `BACKGROUND ---->` $\equiv$ `DEFAULT::BACKGROUND ---->` .

- Namespace separator inside the token is `::` (no surrounding spaces):
`PREFIX::ACTION_NAME` . This is DISTINCT from the action-body separator `::` (spaces both sides) used in forms 1/2 between the token and `<rest>` , and the arrow-body separator `--->` (spaces both sides) used in form 5.

Grammar (extended, anchored to first non-blank line)

- `::` -form regex: `^(?:([A-Z][A-Z0-9_]*)::)?([A-Z][A-Z0-9_]*)::(.*)$`
 $\rightarrow$ group1=optional PREFIX, group2=ACTION_NAME, group3=rest.
- slash-form regex:
`^/(?:([A-Z][A-Z0-9_]*)::)?([A-Z][A-Z0-9_]*)\s+(.*)$` $\rightarrow$
group1=optional PREFIX, group2=ACTION_NAME, group3=rest.
- arrow-form regex: `^(?:([A-Z][A-Z0-9_]*)::)?([A-Z][A-Z0-9_]*) ---> (.*)$` $\rightarrow$ group1=optional PREFIX, group2=ACTION_NAME, group3=rest. The `--->` separator is anchored immediately after the leading token; a mid-prose `--->` after a colon token (e.g. `A :: B ---> C`) parses as the COLON form (the arrow match falls through), so the `::` , arrow, and slash forms never cross-interfere.
- Token case: ACTION_NAME and PREFIX are UPPERCASE
`[A-Z][A-Z0-9_]*` (directive/macro convention — DISTINCT from the §11.4.29 lowercase file/dir naming, which still governs the registry FILE names + script names). `DEFAULT` is the reserved default-namespace literal.
- Escape: leading `\` makes the line literal (no expansion): `\BACKGROUND :: x` , `\BACKGROUND ---> x` , and `\BACKGROUND x` .
- Unknown grammar-shaped token (looks like an action but not in the registry) $\rightarrow$ ask per §11.4.66/§11.4.105, NEVER silently expand/drop (§11.4.6).

Conflict rule (slash form)

- `/ACTION_NAME` (form 3) is honored as the action ONLY when `ACTION_NAME` (case-folded for the collision check) does not match a built-in/host slash command. The registry MAY carry a per-action `slash_bare: true|false|auto` and a `slash_conflicts: [...] list`; `auto` = enable bare slash unless a known host command collides.
- Form 4 (`/PREFIX: :ACTION_NAME`) is ALWAYS unambiguous and always honored.

Registry schema additions

```
grammar:
  default_namespace: DEFAULT
  namespace_separator: ':'      # inside the token, no spaces
  body_separator: ' :: '      # token <-> rest, spaces both sides ('::' form)
  arrow_body_separator: ' ---> ' # token <-> rest, spaces both sides (arrow form)
  slash_prefix: '/'
  slash_body_separator: ' '    # token <-> rest (slash form)
  # regexes: colon_form_regex / slash_form_regex / arrow_form_regex
actions:
  - name: BACKGROUND
    namespaces: [DEFAULT]      # which namespaces this action is registered under
    slash_bare: auto           # bare /BACKGROUND honored unless host-command collision
    slash_conflicts: []        # known colliding host slash commands (forces /
                              # DEFAULT::BACKGROUND)
    # ... expansion, rules, composes_with as before
  - name: REMINDER
    namespaces: [DEFAULT]      # second registered action (verify-don't-assume status re-
                              # surfacing)
    slash_bare: auto
    slash_conflicts: []
    # ... expansion, rules, composes_with
```

Impl impact

- `apx_parse_prefix` MUST recognize all 5 forms, returning (namespace|DEFAULT, action, rest, matched_form ∈ {colon, slash, arrow}). The python and awk parse paths stay byte-identical; the awk arrow branch FALLS THROUGH to the colon check on an invalid leading token so `A :: B ---> C` parses identically in both paths.
- `apx_expand_prompt` resolves namespace+action against the registry; bare-slash honored per the conflict rule; the arrow form resolves to the SAME action/expansion as the `::` and slash forms.
- LAYER-2 hook handles all 5 first-line forms.
- The slash-command generator already emits `/background` (form 3) per agent; ADD the namespaced `/default::background` (form 4) where the agent's command syntax allows `::` (else a `default-background` alias) + document the collision fallback. The arrow form (5) is a free-form/typed delimiter, not a generated slash command.
- `recognition_instruction.md` teaches the agent all 5 forms + the conflict rule + escape + the `REMINDER` action.

Naming: action/namespace tokens UPPERCASE; FILES/DIRS lowercase snake_case (§11.4.29). DEFAULT is the reserved default namespace literal.